

Nirva — AI Engineer · Design Deep Dive

The round at a glance

Two parts:

- 1. Prep (offline, ~60 min suggested).** We send you this prompt ahead of time. Spend as long as you like — 60 minutes is a guide, not a target. Come with whatever helps you think: notes, sketches, a doc, slides, code. There's nothing to submit beforehand.
- 2. Live deep dive (45 min).** You walk us through what you worked on; we ask questions and dig in together.

There's no single correct answer. The problem is deliberately open. We want to see how you reason about messy real-world signals, what you choose to build, and — just as much — what you choose to leave out. We care far more about your thinking than about a polished deliverable.

About Nirva

Nirva is a jewelry-grade AI wearable that passively listens to your voice through the day and turns it into a private life journal and companion. It is not a productivity tracker. The aim is to make someone feel *witnessed* — to reflect back what they actually lived, help them notice their own patterns, and support them in growing over time. The experience should feel personally witnessed, not algorithmically generated. Privacy is foundational: capture is self-only (the wearer's own voice), and the data belongs to the user.

The data you have to work with

Audio is captured through the day and processed in batches. Each processed segment comes out as a record with up to seven dimensions:

Field	Source	Always present?	Notes
transcript	ASR	Yes	The wearer's words only. Self-only capture — no speaker labels.
emotion	EmoNet (40-class)	No — conditional	Returned only when the top label's confidence ≥ 0.70 (a clear emotional signal). When present, it's the top-3 labels with confidences. Otherwise <code>null</code> .
pitch	Librosa	Yes	e.g. mean F0
loudness	Librosa	Yes	e.g. RMS dB

Field	Source	Always present?	Notes
speaking_rate	Librosa	Yes	e.g. syllables/sec
prosody	Librosa	Yes	composite expressiveness / variation measure
pause_frequency	Librosa	Yes	e.g. pauses/min

So six fields (transcript + the five acoustic features) are always non-null. `emotion` is the only optional field — and in real data, most segments carry no emotion tag at all. Units below are illustrative.

Example record — clear emotional signal:

```
{
  "timestamp": "2026-06-07T09:14:32-07:00",
  "transcript": "Honestly the demo went better than I expected – I'd been dreading it all morning.",
  "emotion": {
    "labels": [
      { "label": "relief", "confidence": 0.82 },
      { "label": "pride", "confidence": 0.71 },
      { "label": "anxiety", "confidence": 0.58 }
    ]
  },
  "pitch": 182.4,
  "loudness": -21.7,
  "speaking_rate": 4.1,
  "prosody": 0.63,
  "pause_frequency": 5
}
```

Example record — no clear emotional signal:

```
{
  "timestamp": "2026-06-07T13:48:05-07:00",
  "transcript": "Need to email the team about the BOM and grab groceries on the way home.",
  "emotion": null,
  "pitch": 165.9,
  "loudness": -24.3,
  "speaking_rate": 3.7,
  "prosody": 0.38,
  "pause_frequency": 9
}
```

A single day produces many such records, and the system accumulates them over weeks and months.

Your task

Read the data, understand the format, and — with Nirva's product goal in mind — design the two things below.

- 1. Design how this becomes something meaningful for the user.** Given Nirva's mission, how would you turn this raw, multi-dimensional data into an experience that feels meaningful — and how would you present it? Cover both registers: what the user encounters **every day**, and what emerges only over the **long term** (weeks, months). Be concrete about what each piece is, how it's shaped, and why it earns its place.
- 2. Design the memory & context layer.** How would you store, organize, and retrieve everything the system has heard, so it can produce the experience in task 1 — today, and a year from now? Talk through the structure, what you keep versus drop, and how context gets assembled when it's time to generate.

Bring whatever artifacts help you tell the story. We'll spend most of the session inside your reasoning and tradeoffs.